



Review

Steroid-responsive encephalopathy associated with autoimmune thyroiditis (SREAT): Characteristics, treatment and outcome in 251 cases from the literature☆



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ABSTRACT

Background: Steroid-responsive encephalopathy and associated autoimmune thyroiditis (SREAT) is characterized by encephalopathy and the presence of antithyroid antibodies. We describe the clinical presentation, outcome and treatments for SREAT by a systematic review of the literature.

Methods: MEDLINE via PubMed, Web of Science and the Cochrane Library were searched for articles published until 2015. Inclusion criteria were unexplained encephalopathy with antithyroid antibodies.

Results: We found reports of 251 patients (median age 52 years [range 18–86], 73% females, 80 [32%] with preexisting thyroiditis). Patients presented encephalitis signs with convulsions ($n = 117$; 47%), confusion ($n = 115$, 46%), speech disorder ($n = 91$, 37%), memory impairment ($n = 107$, 43%), gait disturbance ($n = 67$, 27%) and persecutory delusions ($n = 61$, 25%). Twenty-eight patients (11%) presented progressive memory impairment and 26 (10%) isolated psychiatric disorders. In serum, 34% of patients were positive for anti-thyroid peroxidase (TPO) antibodies, 7% for anti-thyroglobulin (TG) antibodies, and 69% both. Thyroid-stimulating hormone levels were usually normal, at 2 UI/ml [0.001–205]. Cerebrospinal fluid from 10/53 patients (19%) was positive for anti-TPO antibodies, 2/53 (4%) anti-TG antibodies and 28 (53%) both. Electroencephalography findings were abnormal for 82% of patients, showing diffuse slowing consistent with encephalopathy (70%) or epileptic activity (14%). The first-line treatment was steroids in 193 patients and other immunosuppressive drugs in 10 cases. At a median follow-up of 12 months [range 0.2–110], 91% of patients showed complete or partial neurological response, with anti-TPO and -TG antibody titers at 347 UI/ml [0–825,000] and 110 UI/ml [0–50,892], respectively. During follow-up, 40 patients (16%) experienced at least one relapse. Relapse was more frequent in patients with initial coma (26% vs 13%, $p = 0.08$).

Conclusion: The diagnosis of SREAT should be suspected in case of encephalopathy without obvious cause, to quickly start corticosteroid treatment. The exact modalities of treatment must be defined.

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1. Introduction

Steroid-responsive encephalopathy associated with autoimmune thyroiditis (SREAT) is a rare disorder first reported by Brain et al., in 1966, as Hashimoto's encephalopathy. SREAT is characterized by encephalopathy associated with antithyroid (anti-thyroid peroxidase [TPO]) and/or anti-thyroglobulin (TG) antibodies, in the absence of alternative causes [1]. Whether this entity is considered encephalopathy, defined as symptoms consistent with neurological and psychiatric signs with the presence of antithyroid antibodies, or real encephalitis related to pathogenic antibodies, is still controversial.

The prevalence of Hashimoto's encephalopathy is not well established and remains underdiagnosed. The clinical presentation varies, from minor cognitive impairment to status epilepticus, for a challenging diagnosis. Usually steroids alone are the first-line treatment, but the initial dose, treatment duration and factors associated with treatment failure remain to be established. Even though steroid dependence and a high rate of relapse have been described, the place of steroid-sparing treatments remains to be determined.

With this literature review, we aimed to describe the characteristics, treatments and outcome of SREAT and analyze the factors associated with treatment response and relapse.

2. Patients and methods

2.1. Literature search strategy

Two investigators (CL and AM) searched MEDLINE via PubMed, Web of Science and the Cochrane Library for English-language articles published up to January 2015 by using the keywords "encephalitis, encephalopathy, steroid-responsive encephalopathy, Hashimoto encephalopathy, neurological signs, antithyroid (anti-TPO and/or anti-TG) antibodies, and thyroiditis". All articles with sufficient data were included in the literature review [7–35].

We included articles describing cases with (1) the presence of signs compatible with encephalopathy, (2) positive antithyroid antibodies (anti-TPO and/or anti-TG antibodies) in serum and/or cerebrospinal fluid; (3) age > 18 years; and (4) absence of other etiologies for encephalitis, considering brain imaging (MRI or tomodesitometry), lumbar puncture, and the absence of autoimmune disease and antinuclear antibodies, Sjögren's syndrome A antigen (anti-SSA) and/or anti-SSB antibodies, or antiphospholipid antibodies.

Clinical, radiological, electroencephalography (EEG) and biological data were extracted from reports. Data on treatments concerned thyroxine substitution, anticonvulsant and anxiolytic drugs, steroids and other immunosuppressive agent doses, duration and treatment indications. Neurological response was defined as complete (disappearance of all symptoms as compared with baseline), partial (at least 50% improvement), and no response for the remaining cases. The data were extracted for cases at diagnosis and at 1, 3, 6 months and the last visit if available.

2.2. Statistical analysis

Data are expressed as median (range) for continuous variables and frequency (%) for qualitative variables. To consider missing data in the analysis, results were expressed relative to the total number of available data. Fisher's exact test was used to compare categorical variables and non-parametric Mann–Whitney or Wilcoxon test to compare continuous

variables. $P < 0.05$ was considered statistically significant. Statistical analyses involved use of GraphPad Prism 5.1 (GraphPad Software, San Diego, USA).

3. Results

Overall, 321 articles were published during the study period; 33 articles were non-English publications and 37 cases concerned patients <18 years old. We included reports of 251 cases in the review.

3.1. Patient characteristics

The median age of patients was 52 years [range 18–86]; 73% were female (Table 1). The initial clinical presentation consisted of convulsions ($n = 117$; 47%), confusion ($n = 115$, 46%), speech disorder ($n = 91$, 37%), memory impairment ($n = 107$, 43%), gait disturbance ($n = 67$, 27%), persecutory delusions ($n = 61$, 25%), myoclonus ($n = 50$, 22%), headaches ($n = 38$, 16%), coma ($n = 38$, 15%) and depression ($n = 27$, 12%). Overall, 28 patients (11%) had isolated progressive memory impairment, with median Mini-Mental State Examination score 21 [3–30], and 26 (10%) had isolated psychiatric disorder (paranoid delusions, depression).

The median thyroid-stimulating hormone (TSH) level was normal, 2 UI/ml [0.001–205]. In all, 80/248 patients (32%) had preexisting disease, 62 with hypothyroidism (Hashimoto's thyroiditis) and 18 hyperthyroidism (Grave's disease).

Serum titers of antithyroid antibodies were increased in all patients. In serum, 119/173 (69%) patients were positive for both anti-TPO and anti-TG antibodies; 70 (34%) only anti-TPO antibodies and 11 (7%) only anti-TG antibodies. Median anti-TPO level at diagnosis was 900 UI/ml [0–270,000] and anti-TG level 179 UI/ml [0–36,569]. In

Table 1

General characteristics of 251 patients with steroid-responsive encephalopathy associated with autoimmune thyroiditis (SREAT).

Characteristics	
Age (years), median [range]/female sex	52 [18–86]/184/251 (73)
Coma	38/249 (15)
Headaches	38/245 (16)
Convulsions	117/250 (47)
Speech disorder	91/249 (37)
Memory impairment	107/247 (43)
Confusion	115/249 (46)
Gait disturbance	67/249 (27)
Persecutory delusions	61/247 (25)
Myoclonus	50/226 (22)
Depression	29/244 (12)
Fever	18/244 (7)
Abnormal electroencephalography findings	182/224 (81)
Serum positivity	
Anti-TPO antibodies	70/203 (34)
Anti-TG antibodies	11/168 (7)
Anti-TGO + TG antibodies	119/173 (69)
Cerebrospinal fluid positivity	
Anti-TPO antibodies	10/53 (19)
Anti-TG antibodies	2/53 (4)
Anti-TPO + TG antibodies	28/53 (53)
TSH, median [range]	2 [0.001–205]
Relapse	40/251 (16)
Death	9/141 (6)
Follow-up (months), median [range]	12 [0.2–110]

Data are no. (%) unless indicated.

cerebrospinal fluid, 10/53 patients (19%) were positive for anti-TPO antibodies, 2/53 (4%) for anti-TG antibodies, and 28 (53%) both, and positivity was always associated with positivity for antithyroid antibodies in serum.

Cerebrospinal fluid analysis showed mild to moderate hyperproteinorachia in 107/131 patients (82%), with median level 0.71 g/l [0.13–7.65]. Pleiocytosis was noted in 26/133 patients (20%) with median level 8 elements/mm³ [1–552] and only 3/133 patients (2%) had more than 60 elements/mm³. All 51 patients with available data were negative for antineuronal antibodies (mostly anti-Hu and anti-Yo antibodies).

EEG findings were abnormal for 182/224 patients (82%) and showed diffuse slowing without epileptic activity consistent with encephalopathy in 129/184 cases (70%). In 25/184 patients (14%), EEG showed epileptic activity consistent with seizures. MRI was performed for 230 patients and findings were abnormal for 119 (52%), showing non-specific white-matter hyperintensities.

As compared with non-coma patients at diagnosis, those with coma ($n = 38$) were more frequently female (89% vs 71%, $p = 0.02$) and had a higher rate of convulsions (84% vs 40%, $p < 0.0001$) (Table 2). Patients with anti-TPO and/or anti-TG antibodies had similar initial clinical presentation, EEG and MRI findings, and only the rate of relapse during the follow-up seemed to differ (0 vs 13% and 18% for patients with anti-TPO and/or anti-TG antibodies).

3.2. Treatments and outcome

Among 228 patients with sufficient data, 25 refused steroid treatment, and finally 203 received treatment at SREAT diagnosis (Fig. 1); 193/203 (95%) received steroids alone as first-line treatment, preceded by intravenous infusion in 150 cases ($1 \text{ g} \times 3$ days in 65 cases), then oral prednisone at a median of 60 mg/day [10–150].

Ten patients (5%) received other immunosuppressive drugs at SREAT diagnosis: intravenous immunoglobulins ($n = 4$), azathioprine ($n = 2$), plasma exchanges ($n = 2$), rituximab ($n = 1$) or hydroxychloroquine ($n = 1$), combined with steroids in 5 cases. Patients who received steroids or other immunosuppressive drugs at diagnosis did not differ in clinical, biological, or radiological features, except for presence of initial coma (16% for patients who received steroids alone vs none).

Table 2
Comparison of patients with coma and without coma at SREAT diagnosis.

	Coma ($n = 38$)	No coma ($n = 211$)
Age, median [range]	51 [21–83]	52 [18–86]
Sex female	34/38 (89)	149/211 (71)*
Convulsions	32/38 (84)	83/210 (40)*
Speech disorder	14/36 (39)	77/211 (36)
Gait disturbance	9/37 (24)	58/210 (28)
Myoclonus	11/33 (33)	39/191 (20)
Confusion	23/37 (62)	92/210 (44)
Persecutory delusions	12/37 (32)	49/208 (24)
Fever	6/36 (17)	11/206 (5)*
Abnormal electroencephalography findings	34/34 (100)	147/188 (78)*
Plasma titers (UI/ml)		
Anti-TPO antibody	640 [0–15,684]	933 [0–270,000]
Anti-TG antibody	284 [0–36,569]	142 [0–29,043]
Proteinorachia (mg/l)	68 [25–228]	73 [13–765]
TSH (UI/ml)	2 [0.03–100]	2 [0.01–205]
Steroids	32/36 (89)	162/187 (86)
Other immunosuppressive agents	6/31 (19)	33/177 (19)
Response to therapy		
First visit: complete	5/27 (19)	20/116 (17)
First visit: partial	20/27 (74)	87/116 (75)
Second visit: complete	4/12 (33)	23/55 (42)
Second visit: partial	3/12 (25)	25/55 (45)

Data are no. (%) unless indicated.

* $p < 0.05$.

Thyroxine substitution was used in 35/132 (27%) patients for hypothyroidism, at a median dose of 100 µg/day [25–200]. Antiepileptic and anxiolytic agents were used in 73/173 (42%) and 30/144 cases (21%), respectively.

Overall, 59 patients (24%) received second-line treatment because of insufficient response ($n = 14$), relapse ($n = 5$), steroid dependence ($n = 11$) or adverse events ($n = 3$). Treatments consisted of steroids alone ($n = 20$), other immunosuppressive treatment without steroids ($n = 33$), and steroids combined with other drugs ($n = 6$). Other immunosuppressive treatments were azathioprine ($n = 15$), plasma exchanges ($n = 9$), intravenous immunoglobulins ($n = 6$), methotrexate ($n = 3$), rituximab ($n = 2$), cyclophosphamide ($n = 2$) and mycophenolate mofetil ($n = 2$).

A complete or partial neurological response to treatment occurred in 146/157 (93%) cases within the first 3 months. Among 146 patients evaluated within a median delay of 5 days [1–20], 134/146 (92%) showed neurological response (complete or partial); in this group, TPO antibody titers decreased from 733 UI/ml [0–25,600] at diagnosis to 293 UI/ml [0–4200] at the first visit ($p < 0.05$). Among 70 patients evaluated within 30 days [10–90], 81% had persistent neurological response, with TPO and TG antibody titers at 150 UI/ml [1–1924] and 374 UI/ml [1–3000], respectively. Overall, 86% of patients were evaluated at 90 days [30–730] and 77% showed persistent neurological response.

At a median follow-up of 12 months [0.2–110], 168/184 patients (91%) showed complete or partial neurological response, with anti-TPO and anti-TG antibody titers at 347 UI/ml [0–825,000] and 110 UI/ml [0–50,892], respectively. Among 70 patients with sufficient data, 36 were still receiving steroids at the last visit, with median dose of 10 mg/day [2.5–60].

Complete responders ($n = 25$) and partial or non-responders ($n = 119$) at the first visit did not differ in baseline characteristics, time from first symptoms to treatment, clinical presentation or antithyroid antibody titers. The rate of complete response was similar between patients with intravenous steroids and oral prednisone at SREAT diagnosis, at the first visit (20% and 17%, respectively) and at the second visit (both 38%) and had similar relapse rates (16% and 14%, respectively).

During a median follow-up of 12 months [0.2–110], 40 patients (16%) experienced at least one relapse. The presence of initial coma was more frequent in patients with relapse (26% vs 13%, $p = 0.08$) and the relapse was less associated with positivity for anti-TG than anti-TPO antibodies or both (0 vs 13% and 18%, respectively).

Nine of 141 patients (6%) died during follow-up and the death was due to infection ($n = 2$), cardiac infarction and myocarditis ($n = 1$ each) or remained unexplained.

4. Discussion

In our study of 251 cases, the largest series of SREAT as far as we know, SREAT could have various presentations, from coma to progressive cognitive impairment and isolated psychiatric disorder. MRI and cerebrospinal fluid findings could be normal or show non-specific features, so antithyroid antibodies should be tested in serum and cerebrospinal fluid in all patients with encephalopathy without obvious etiology. Treatment with steroids was highly effective: more than 90% of the cases showed persistent neurological improvement, and the oral prednisone was as effective as intravenous steroids.

SREAT could present as encephalopathy and could feature psychiatric disorder or progressive cognitive impairment. Castillo et al. [2] found cognitive impairment in 20 cases of SREAT, seizures in 60%, and psychiatric presentation, psychosis or paranoia in 25%. The initial clinical diagnoses among these 20 cases varied: infectious encephalitis, Creutzfeldt–Jacob disease, stroke transient ischemic attack or Alzheimer disease. One of the challenging differential diagnoses is paraneoplastic limbic encephalitis. For 24 patients with paraneoplastic limbic encephalitis, the clinical signs were cognitive dysfunction, psychiatric

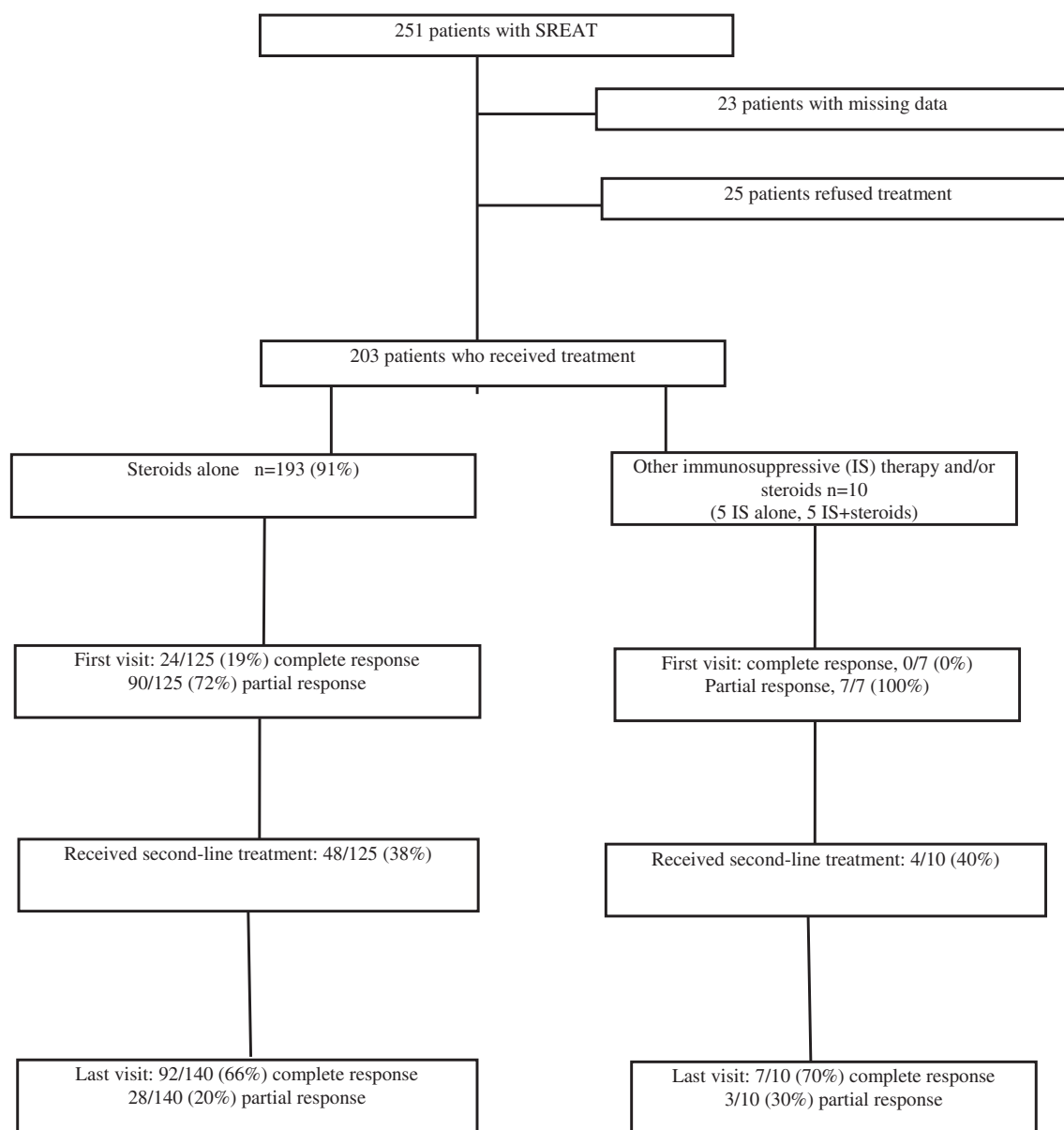


Fig. 1. Flow chart.

symptoms and seizures; in most patients, neurological symptoms preceded the diagnosis of cancer by a median of 6 months, with antineuronal antibodies detected in 75% of patients [3]. One of the other main differential diagnoses is NMDA receptor encephalitis. In our study, antineuronal antibodies, including NMDA antibodies, were searched in less than 20% of cases. Dalmau et al. reported 577 patients with NMDA receptor encephalitis [4]; the median age of 21 years was much younger than for our SREAT cases (52 years), and the main presentation was behavioral changes in 65% of cases. Patients had worst presentation than did our SREAT cases, with 77% admitted to an intensive care unit.

Only 32% of our patients had known thyroid disease and most had normal TSH levels, so antithyroid antibodies should be systematically tested in all patients with unexplained encephalopathy. Because no clinical features or EEG or MRI signs are specific for SREAT, antithyroid antibodies are one of the keys to diagnosis. Nevertheless, antithyroid antibodies in serum are frequent: they could be present in 13% of healthy individuals and 27% of women >60 years old [6]; with median age in our study at 52 years and predominant female sex. The presence

of anti-thyroid antibodies in cerebrospinal fluid could be a strong argument for the disease, but the prevalence of these antibodies in the general population is unknown and no standardized assay is available. Therefore, the term steroid-responsive encephalopathy and associated autoimmune thyroiditis (SREAT) is appropriate and emphasizes that the pathogenicity of antithyroid antibodies is not clearly demonstrated and SREAT may include other steroid-responsive encephalitis. Hence, the steroid response is another feature that should be included in the final diagnosis of SREAT, even if a few cases might not respond to steroids and need other immunosuppressive agents.

In our study, 89% of the patients (203/228) received immunosuppressive treatment as first-line therapy, mainly steroids alone. Steroid treatment was highly effective, and 91% of patients showed partial or complete neurological response. Less than 5% received therapy other than steroids or associated immunosuppressive agents as first-line treatment, mainly for steroids contraindications. The benefit and the best regimen maintenance treatment remains to be determined, and various drugs, such as disease-modifying antirheumatic drugs, immunoglobulins and rituximab, have been used. Intravenous steroids

and oral prednisone have similar response rates to that with other immunosuppressive drugs associated with steroids at SREAT diagnosis, which argues for oral steroids at least in non-complicated cases.

Overall, 16% of our patients had at least one relapse, and relapse was more frequent in patients with initial coma, whereas anti-TG antibody positivity alone seemed to be associated with good outcome. Previously, high serum anti-TPO antibody titers were found associated with favorable outcome and increased age with worse outcome [5].

5. Conclusion

SREAT is a rare syndrome with various clinical presentations that should be considered in any patient with acute or subacute encephalopathy, with progressive cognitive impairment and psychiatric signs without obvious etiology. SREAT belongs to steroid-responsive encephalitis, so steroids remain the first-line treatment and must be started quickly. Most of the time, treatment leads to marked improvement, but the benefit of other immunosuppressive drugs, in particular in cases with insufficient response to steroids or relapse, remains to be determined.

Take-home messages

- SREAT could have various presentations, from coma to progressive cognitive impairment and isolated psychiatric disorder.
- MRI and cerebrospinal fluid findings could be normal or show non-specific features, so antithyroid antibodies should be tested in serum and cerebrospinal fluid in all patients with encephalopathy without obvious etiology.
- Treatment with steroids is highly effective and the oral prednisone seem to be as effective as intravenous steroids.

References

- [1] Graus F, Titulaer MJ, Balu R, Benseler S, Bien CG, Cellucci T, et al. A clinical approach to diagnosis of autoimmune encephalitis. *Lancet Neurol* 2016;15(4):391–404.
- [2] Castillo P, Woodruff B, Caselli R, Vernino S, Lucchinetti C, Swanson J, et al. Steroid-responsive encephalopathy associated with autoimmune thyroiditis. *Arch Neurol* Feb 2006;63(2):197–202.
- [3] Lawn ND, Westmoreland BF, Kiely MJ, Lennon VA, Vernino S. Clinical, magnetic resonance imaging, and electroencephalographic findings in paraneoplastic limbic encephalitis. *Mayo Clin Proc* 2003;78:1363–8.
- [4] Titulaer MJ, McCracken L, Gabilonda I, Armangué T, Glaser C, Lizuka T, et al. Treatment and prognostic factors for long-term outcome in patients with anti NMDA receptor encephalitis: an observational cohort study. *Lancet Neurol* Feb 2013;12(2):157–65.
- [5] Litmeier S, Prüss H, Witsch E, Witsch J. Initial serum thyroid peroxidase antibodies and long-term outcomes in SREAT. *Acta Neurol Scand* Jan 12 2016.
- [6] Hollowell JG, Staehling NW, Flanders WD, Hannon WH, Gunter EW, Spencer CA, et al. Serum TSH, T(4), and thyroid antibodies in the United States population (1988 to 1994): National Health and Nutrition Examination Survey (NHANES III). *J Clin Endocrinol Metab* Feb 2002;87(2):489–99.
- [7] Florance NR, Davis RL, Lam C, Szperka C, Zhou L, Ahmad S, et al. Anti-N-methyl-D-aspartate receptor (NMDAR) encephalitis in children and adolescents. *Ann Neurol* Jul 2009;66(1):11–8.
- [8] Dalmau J, Tuzun E, Wu HY, Masjuan J, Rossi JE, Voloschin A, et al. Paraneoplastic anti-N-methyl-D-aspartate receptor encephalitis associated with ovarian teratoma. *Ann Neurol* 2007;61:25–36.
- [9] Olmez I, Moses H, Sriram S, Kirshner H, Lagrange AH, Pawate S. Diagnostic and therapeutic aspects of Hashimoto's encephalopathy. *J Neurol Sci* Aug 15 2013;331(1–2):67–71.
- [10] Zhao W, Li J, Wang J, Guo Y, Tuo H, Kang Z, et al. A case of Hashimoto encephalopathy: clinical manifestation, imaging, pathology, treatment, and prognosis. *Neurologist* May 2011;17(3):141–3.
- [11] Pari E, Rinaldi F, Premi E, Codella M, Rao R, Paghera B, et al. A follow-up 18F-FDG brain PET study in a case of Hashimoto's encephalopathy causing drug-resistant status epilepticus treated with plasmapheresis. *J Neurol* Apr 2014;261(4):663–7.
- [12] Nayak HK, Daga MK, Kumar R, Garg SK, Kumar N, Mohanty PK. A series report of autoimmune hypothyroidism associated with Hashimoto's encephalopathy: an under diagnosed clinical entity with good prognosis. *BMJ Case Rep* Nov 2010;23:2010.
- [13] Caselli RJ, Drazkowski JF, Wingerchuk DM. Autoimmune encephalopathy. *Mayo Clin Proc* Oct 2010;85(10):878–80.
- [14] Henchey R, Cibula J, Helveston W, Malone J, Gilmore RL. Electroencephalographic findings in Hashimoto's encephalopathy. *Neurology* May 1995;45(5):977–81.
- [15] Cantón A, de Fàbregas O, Tintoré M, Mesa J, Codina A, Simó R. Encephalopathy associated to autoimmune thyroid disease: a more appropriate term for an underestimated condition? *J Neurol Sci* May 1 2000;176(1):65–9.
- [16] Kothbauer-Margreiter I, Sturzenegger M, Komor J, Baumgartner R, Hess CW. Encephalopathy associated with Hashimoto thyroiditis: diagnosis and treatment. *J Neurol* Aug 1996;243(8):585–93.
- [17] Archambeaud F, Galinat S, Regoubay Y, Magy L, Rebeyrotte I, Vallat JM, et al. Hashimoto encephalopathy. Analysis of four case reports. *Rev Med Interne* Jul 2001;22(7):653–9.
- [18] Sharma PMS, Javali M, Mahale R, Madhusudhan BK, Majeed AA, Srinivasa R. Hashimoto encephalopathy: a study of the clinical profile, radiological and electrophysiological correlation in a Tertiary Care Center in South India. *J Neurosci Rural Pract* Sep 2015;6(3):309–14.
- [19] Chang T, Riffy MT, Gunaratne PS. Hashimoto encephalopathy: clinical and MRI improvement following high-dose corticosteroid therapy. *Neurologist* Nov 2010;16(6):394–6.
- [20] Chaigne B, Beaufils E, Jouan Y, Magnant J, Maillot F, Constans T, et al. Hashimoto's encephalopathy. *Rev Med Interne* Jul 2012;33(7):390–5.
- [21] Kashyap AS, Kashyap S. Hashimoto's encephalopathy. *Lancet* May 31 2003;361(9372):1912.
- [22] Bhansali A, Jayaprakash P, Dutta P, Walia R, Anantharaman R, Modi M. Hashimoto's encephalopathy: an under diagnosed entity. *BMJ Case Rep* 2009;2009.
- [23] Shaw PJ, Walls TJ, Newman PK, Cleland PG, Cartledge NE. Hashimoto's encephalopathy: a steroid-responsive disorder associated with high anti-thyroid antibody titers—report of 5 cases. *Neurology* Feb 1991;41(2 (Pt 1)):228–33.
- [24] Rekand T, Vedeler C, Gramstad A, Bindoff L. Hashimoto's encephalopathy: a treatable cause of mental impairment, stroke and seizures. *Eur J Neurol* Nov 2003;10(6):746–7.
- [25] Bostantjopoulou S, Zafiriou D, Katsarou Z, Kazis A. Hashimoto's encephalopathy: clinical and laboratory findings. *Funct Neurol* Oct 1996;11(5):247–51.
- [26] Ferracci F, Bertiato G, Moretto G. Hashimoto's encephalopathy: epidemiologic data and pathogenetic considerations. *J Neurol Sci* Feb 15 2004;217(2):165–8.
- [27] Duffey P, Yee S, Reid IN, Bridges LR. Hashimoto's encephalopathy: postmortem findings after fatal status epilepticus. *Neurology* Oct 28 2003;61(8):1124–6.
- [28] Chang J-S, Chang T-C. Hashimoto's encephalopathy: report of three cases. *J Formos Med Assoc* Nov 2014;113(11):862–6.
- [29] Marshall GA, Doyle JJ. Long-term treatment of Hashimoto's encephalopathy. *J Neuropsychiatry Clin Neurosci* 2006;18(1):14–20.
- [30] Spiegel J, Hellwig D, Becker G, Müller M. Progressive dementia caused by Hashimoto's encephalopathy—report of two cases. *Eur J Neurol* Oct 2004;11(10):711–3.
- [31] Sawka AM, Fatourehchi V, Boeve BF, Mokri B. Rarity of encephalopathy associated with autoimmune thyroiditis: a case series from Mayo Clinic from 1950 to 1996. *Thyroid* May 2002;12(5):393–8.
- [32] Bohnen NI, Parnell KJ, Harper CM. Reversible MRI findings in a patient with Hashimoto's encephalopathy. *Neurology* Jul 1997;49(1):246–7.
- [33] Chaudhuri A, Behan PO. The clinical spectrum, diagnosis, pathogenesis and treatment of Hashimoto's encephalopathy (recurrent acute disseminated encephalomyelitis). *Curr Med Chem Oct* 2003;10(19):1945–53.
- [34] Cook MK, Malkin M, Karafin MS. The use of plasma exchange in Hashimoto's encephalopathy: a case report and review of the literature. *J Clin Apher* Jun 2015;30(3):188–92.
- [35] Chong JY, Rowland LP. What's in a NAIM? Hashimoto encephalopathy, steroid-responsive encephalopathy associated with autoimmune thyroiditis, or nonvasculitic autoimmune meningoencephalitis? *Arch Neurol* Feb 2006;63(2):175–6.